

TITANIUM DIOXIDE (NANOFORM) P25**ICSC: 1782 (November 2019)**

Pigmentary titanium dioxide

Particles <100nm

CAS #: 13463-67-7; 1317-70-7; 1317-80-2**EC Number: 215-280-1; 236-675-5; 215-282-2**

	ACUTE HAZARDS	PREVENTION	FIRE FIGHTING
FIRE & EXPLOSION	Not combustible.		In case of fire in the surroundings, use appropriate extinguishing media.

PREVENT DISPERSION OF DUST! Keep exposure as low as reasonably achievable.

	SYMPTOMS	PREVENTION	FIRST AID
Inhalation		Avoid inhalation of fine dust. Use breathing protection.	
Skin		Protective gloves.	Rinse and then wash skin with water and soap.
Eyes	Redness.	Wear safety spectacles.	First rinse with plenty of water for several minutes (remove contact lenses if easily possible), then refer for medical attention.
Ingestion		Do not eat, drink, or smoke during work.	Rinse mouth.

SPILLAGE DISPOSAL	CLASSIFICATION & LABELLING
Personal protection: particulate filter respirator adapted to the airborne concentration of the substance. If appropriate, moisten first to prevent dusting. Sweep spilled substance into covered containers.	According to UN GHS Criteria  Causes damage to the lungs through prolonged or repeated exposure Suspected of causing cancer Transportation UN Classification
STORAGE	
Separated from reducing agents.	
PACKAGING	



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TITANIUM DIOXIDE (NANOFORM) P25**ICSC: 1782****PHYSICAL & CHEMICAL INFORMATION****Physical State; Appearance**
WHITE-TO-YELLOW POWDER.**Physical dangers****Chemical dangers**Formula: TiO₂
Molecular mass: 79.9
Boiling point: 2600-3000°C
Melting point: 1855°C**EXPOSURE & HEALTH EFFECTS****Routes of exposure**

The substance can be absorbed into the body by inhalation and by ingestion.

Effects of short-term exposure**Inhalation risk**

A harmful concentration of airborne particles can be reached quickly when dispersed, especially if powdered.

Effects of long-term or repeated exposure

Lungs may be affected by repeated or prolonged exposure. This substance is possibly carcinogenic to humans if inhaled.

OCCUPATIONAL EXPOSURE LIMITS**ENVIRONMENT**

This substance does enter the environment under normal use. Great care, however, should be taken to avoid any additional release, for example through inappropriate disposal.

NOTES

The CAS numbers for different forms are: CAS 13463-67-7 (anatase/rutile); CAS 1317-70-7 (anatase); CAS 1317-80-2 (rutile). See ICSC 338.

Physical Properties: In addition to the Physical Properties listed in this ICSC, OECD (2016) proposes another 10 to 14 characteristic parameters (for example agglomeration degree, surface chemistry, radical formation potential) in order to sufficiently identify a nanoform.

Additional Properties: Crystallinity: Anatase 83-87%/Rutile 13-17% TEM 1.3-41.9 Zeta: >20mV at pH 5; Photocatalysis: Degradation of Orange II dye in combination with UVB exposure, hydroxyl radical formation with UV radiation; Bioaccumulation: Significant absorption within 30 minutes in fish (C carpio).

ADDITIONAL INFORMATION**EC Classification**

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